

Positive witnesses and a constructive counterfactual system for the Erdős–Straus equation

The Clankers research workbench

Assistant-developed continuation; source-specific credit retained

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Abstract

The same full finite shell atlas supports two exact research directions. Every canonical divisor state gives a positive rational decomposition of $4/p$; the two nondistinguished denominators have the same reduced denominator D . We compute D exactly, identify failure at p with an explicitly invertible finite diagonal operator, and carry both successful and unsuccessful states through marked Cayley, factor, Leech and octonionic coordinates. Integer collision-energy bounds and an exact largest non-subgroup-divisor calculation supply constructive sufficient conditions for $D = 1$, including a strict improvement on the preceding angular bound. No counterexample is found and no universal positivity theorem is claimed. The all-shell criterion itself, the E/M parametrizations, and the standard inequalities are inherited; the present contribution is the explicit two-sided continuation and its certificates, without a historical-priority assertion.

1 Scope, source lineage, and the Giuga model

The target is $4/n = 1/x + 1/y + 1/z$ with positive integer denominators. A counterexample to a restricted ray family, an angular chamber, or the cutoff $R < p$ is not a counterexample to this equation. We use the entire first-denominator range forced by a smallest denominator. The foundation’s `all_shell_no_hit.tex`, pinned at `9a3ddfe8b38a424f1ff0426201a1f4f6f9d2772f`, already supplies the finite exponent packet and complete ordered return [7]. The antecedents are Yamamoto’s Lemma 2, equation (4), p. 38; Monks–Velingker’s reduced-fraction criterion, Theorem 2.2, pp. 4–6; and Elsholtz–Tao’s Type I/II maps, §2, Propositions 2.2 and 2.6 [2, 3, 4]. All formulas required here are rederived below.

The uploaded `verify_giuga_blind_plane_cd.py` separates a Giuga integer from a counterexample to Giuga’s conjecture. Its blocks A3–A6 retain the additional Carmichael conditions. For a composite candidate n and each prime $q \mid n$, the simultaneous local conditions are

$$q \mid n/q - 1, \quad q - 1 \mid n - 1 \iff q^2(q - 1) \mid n - q.$$

The equivalence follows from $q^2 \mid n - q$ and $n - q \equiv n - 1 \pmod{q - 1}$, since $\gcd(q^2, q - 1) = 1$. The first condition forces squarefreeness. The counterexample must be both Giuga and Carmichael, not merely one of them; see Kellner’s Theorem 3.5 [5]. We import this *method of retaining simultaneous conditions*, not a conjectural map between Giuga and ES counterexamples. The file’s A7 uses floating complex exponentials, while A9b checks an exact CRT bijection; neither is represented here as a full formal proof.

For exact phase conventions, the pair $(e^{2\pi i A/q}, e^{2\pi i B/(q-1)})$ has relative exponent

$$e = (q - 1)A - qB \pmod{q(q - 1)}.$$

Its inverse is $A = -e \pmod{q}$, $B = -e \pmod{q - 1}$. Thus this restricted relative phase is faithful, although a generic equal-radius Hopf quotient forgets the common phase. This fixes the signs explicitly without relying on numerical equality of roots of unity. The source’s generic annihilator and Cayley–Dickson experiments are not needed for the ES proof below.

2 The full candidate atlas

Fix a prime $p = 4m + 1$, $m \geq 1$, and set

$$\mathcal{S}_p = \{m + 1, \dots, 3m\}, \quad R_a = 4a - p.$$

Then $3 \leq R_a \leq 2p - 3$, $R_a \equiv 3 \pmod{4}$, and $\gcd(a, R_a) = \gcd(p, R_a) = 1$. These follow from $0 < a < p$ and primality. Write $a = \prod_q q^{e_q}$ and retain the whole box

$$\mathcal{B}_a = \prod_{q|a} \{0, \dots, 2e_q\}, \quad u(f) = \prod_{q|a} q^{f_q}.$$

Unique factorization makes this a bijection to $\text{Div}(a^2)$, with inverse $f_q = v_q(u)$. No generated subgroup replaces this box.

A canonical candidate is $\alpha = (p, a, u, c)$ with $a \in \mathcal{S}_p$, $u \mid a^2$, and $c \in \{E, M\}$. Define

$$Q_E(\alpha) = \left(a, \frac{pa + a^2/u}{R_a}, \frac{pa + p^2u}{R_a} \right), \quad (1)$$

$$Q_M(\alpha) = \left(a, \frac{p(a + a^2/u)}{R_a}, \frac{p(a + u)}{R_a} \right). \quad (2)$$

The initial position is distinguished and the other two positions remain ordered. An additional exterior swap bit is used when all raw orders are required. A middle swap is already $u \mapsto a^2/u$; adding another middle swap bit would duplicate records.

Theorem 2.1 (Positive rational atlas and complete integral test). *Every candidate in (1)–(2) is a positive rational solution of $4/p = 1/a + 1/Y + 1/Z$. With*

$$g = \gcd(a, u), \quad r = u/g, \quad s = a/g, \quad h = g^2/u,$$

one has $h, r, s \in \mathbb{Z}_{>0}$, $\gcd(r, s) = 1$, $a = hrs$, and $u = hr^2$. The rational quotients are

$$\kappa = (pr + s)/R_a \quad (E), \quad \lambda = (r + s)/R_a \quad (M).$$

The ordered maps are $(hrs, hsk, phr\kappa)$ and $(hrs, phs\lambda, phr\lambda)$. Moreover,

$$\text{ES}(p) \iff \text{at least one canonical candidate has all coordinates integral.}$$

There are at most $(p - 1)^2$ canonical candidates.

Proof. At a prime with $v_q(a) = e$ and $v_q(u) = f$, the exponents of r, s, h are $(f - e)_+, (e - f)_+$ and $e - |f - e|$. They are nonnegative and prove all normalization assertions. Direct substitution gives

$$(R_a Y - pa)(R_a Z - pa) = p^2 a^2$$

in both channels. Expanding yields $R_a YZ = pa(Y + Z)$, the desired identity; every displayed denominator is positive. The quotient formulas follow by substitution.

For completeness take an actual integral solution and move its smallest denominator to the first position, retaining that permutation. Positivity gives $p/4 < a \leq 3p/4$, hence $a \in \mathcal{S}_p$. Put $d_Y = R_a Y - pa$, $d_Z = R_a Z - pa$. Then $d_Y, d_Z > 0$ and $d_Y d_Z = p^2 a^2$. Thus $d_Y = p^i v$ for $i \in \{0, 1, 2\}$, $v \mid a^2$. If $i = 0$, the inverse is $E, u = a^2/d_Y$ without a swap; if $i = 1$, it is $M, u = pa^2/d_Y$; if $i = 2$, it is $E, u = d_Y/p^2$ with the exterior swap. The congruences $d_Y \equiv d_Z \equiv -pa \pmod{R_a}$ give exactly the required channel gates. For instance $i = 0$ gives $R_a \mid 4u + 1$ after multiplication by units u, a^{-1} and use of $p \equiv 4a$; $i = 1$ gives $R_a \mid u + a$. Conversely an integral candidate is already a witness. The explicit inverse also preserves the full divisor exponent vector.

Pair divisors of a^2 around its square root a to get $\tau(a^2) \leq 2a - 1$. Hence

$$|\mathcal{A}_p| = 2 \sum_{a=m+1}^{3m} \tau(a^2) \leq 2 \sum_{a=m+1}^{3m} (2a - 1) = 16m^2 = (p - 1)^2.$$

The bound counts canonical candidates, not their additional exterior swaps. $\square$

3 The exact return denominator

Theorem 3.1 (Common denominator and defect). *The two nondistinguished coordinates of each candidate have the same reduced rational denominator*

$$D_E = \frac{R_a}{\gcd(R_a, 4u+1)}, \quad D_M = \frac{R_a}{\gcd(R_a, u+a)}. \quad (\text{D})$$

This D is the least positive integer clearing the entire triple. It is odd, divides R_a , and is prime to p . With

$$X = DQ_c(\alpha), \quad n = pD, \quad \delta_\alpha = (D-1)/2,$$

one has $X \in \mathbb{Z}_{>0}^3$, $\sum 1/X_i = 4/n$, and

$$F_p(X) := 4X_0X_1X_2 - p \sum_{i<j} X_iX_j = 2p\delta_\alpha \sum_{i<j} X_iX_j. \quad (\text{F})$$

Success at the original prime is exactly $D = 1$. If $D > 1$, then $3p \leq n \leq p(2p-3)$ and $1 \leq \delta_\alpha \leq p-2$.

Proof. For E , multiply the numerator of Y by the unit u modulo R_a :

$$u(pa + a^2/u) \equiv a^2(4u+1), \quad pa + p^2u \equiv 4a^2(4u+1) \pmod{R_a}.$$

Every displayed multiplier $u, a, 4$ is a unit. Thus both numerator gcds with R_a equal $\gcd(R_a, 4u+1)$. For M , the numerator identities

$$up(a + a^2/u) = pa(u+a), \quad p(a+u) = p(u+a)$$

give the same conclusion with $u+a$. Reducing the fractions proves (D). As a is integral, D is the least clearing factor of all three entries. The bounds and coprimality follow from the shell units. Finally $F_{pD}(X) = 0$ by scaling the reciprocal identity; subtracting it from $F_p(X)$ gives (F). $\square$

Proposition 3.2 (Complete candidate inverse and collision stratum). *In the canonical order,*

$$u = \frac{a^2}{R_a Y - pa} \ (E), \quad u = \frac{pa^2}{R_a Y - pa} \ (M).$$

An exterior triple has three distinct entries. A middle triple has a repeat precisely when $u = a$, in which case $Y = Z$ and $D = R_a > 1$. The middle complement preserves D and reverses the last two entries.

Proof. The residual products in the two channels are respectively $(a^2/u, p^2u)$ and $(pa^2/u, pu)$, proving the inverse. In the exterior channel $v_p(Y) = 0$ and $v_p(Z) \geq 1$; $a = Y$ would give $(4u-1)a = 2pu$. Since $\gcd(u, 4u-1) = 1$, $4u-1$ divides $2p$, impossible modulo four for its odd divisor. The equality $a = Z$ would imply $p \mid 4a^2$, also impossible. In the middle channel both Y, Z have positive p -adic valuation, while a does not. The equality $Y = Z$ is $a^2/u = u$, hence $u = a$. Then $\gcd(R_a, 2a) = 1$ gives $D = R_a$. Complementation follows from the displayed formulas. $\square$

Proposition 3.3 (A same-shell map cannot improve this defect). *If $u, u' \mid a^2$ and $u' \equiv pu \pmod{R_a}$, the exterior candidate at u and the middle candidate at u' have exactly the same D , whether or not they are integral.*

Proof. $u' + a \equiv a(4u+1) \pmod{R_a}$, and a is a unit. Apply (D). $\square$

This extends the inherited same-shell congruence to unsuccessful rational states. It does not assert that a suitable u' always exists, nor does failure of this particular improvement mechanism rule out other maps.

4 The counterfactual object and its finite inverse certificate

Definition 4.1. The *defect atlas* at p consists of all shells, their complete prime factorizations and exponent boxes, both canonical channel maps, each D , and each local valuation deficit

$$v_\ell(D) = \max\{v_\ell(R_a) - v_\ell(g_c(u)), 0\}, \quad g_E(u) = 4u + 1, \quad g_M(u) = u + a.$$

A negative certificate lists every canonical candidate and, for each one, a prime ℓ where the displayed deficit is positive. It has the explicit scope *full* or *restricted*.

Let $\mathcal{A}_p = \mathbb{Q}^{\mathcal{A}_p}$ be the finite algebra of functions on the canonical candidate set, retaining its basis labels. Let H_p be multiplication by $\delta_\alpha = (D_\alpha - 1)/2$.

Theorem 4.2 (Two-sided certificate theorem). *The following are equivalent:*

1. p has no positive integral ES decomposition;
2. $D_\alpha \geq 3$ for every canonical candidate;
3. H_p is invertible;
4. there is a unique function b with $\delta b = 1$, namely $b(\alpha) = 2/(D_\alpha - 1)$;
5. there is a full negative certificate with the stated local deficits.

When these conditions hold, $0 < b(\alpha) \leq 1$. Otherwise any zero coordinate of H_p reconstructs a positive integral witness. Both alternatives are recognizable by terminating exact algorithms.

Proof. The atlas theorem and the common denominator theorem identify existence with some $D = 1$. Every other D is odd and at least three. A diagonal operator on a finite coordinate algebra is invertible exactly when all entries are nonzero, with the displayed unique inverse. A nontrivial positive integer D has a prime divisor, whose valuation has the stated positive deficit. Conversely any such deficit proves $D > 1$. Enumerating the finite shell interval and all bounded exponent choices constructs one of the alternatives and verifies it. Primality is checked by trial division through $\lfloor \sqrt{p} \rfloor$ in the supplied verifier. $\square$

There is a universal polynomial version, not an uncomputed hypothesis. Set

$$\Pi_p(z) = \prod_{j=1}^{p-2} (1 - z/j), \quad Q_p(z) = \frac{1 - \Pi_p(z)}{z} \in \mathbb{Q}[z].$$

All diagonal entries of H_p belong to $\{0, \dots, p-2\}$, so

$$\Pi_p(H_p) = \text{the exact success projector}, \quad H_p Q_p(H_p) = I - \Pi_p(H_p).$$

Thus a false prime has $H_p Q_p(H_p) = I$. The success dimension is the number of canonical passing candidates, not a count of globally sorted triples. This interpolation is a standard finite-algebra fact; the particular operator and its arithmetic labels are the construction here.

A genuine restricted prototype, not a counterexample. At $p = 241, a = 64, R = 15$ all divisors are $u = 2^e, 0 \leq e \leq 12$. For either channel the D histogram is

$$D = 3 : 4 \text{ entries}, \quad D = 5 : 6 \text{ entries}, \quad D = 15 : 3 \text{ entries}.$$

The negative operator is invertible on these 26 slots. Yet the *full* atlas at 241 has positive witnesses in other shells. Even $\gcd\{D_\alpha\} = 1$ on this rejected shell: taking a gcd across different candidates would give a false conclusion. The second prototype is $p = 37, a = 18, R = 35$, whose integral

exponent lift can lie outside the original box. Both prototypes are already motivated in the foundation and are replayed here with their full rational defect data.

The same example has a complete defect-preserving transport graph. Its vertices are the 26 labelled E/M candidates. Join complementary middle states, and join $E(u)$ to $M(u')$ precisely when $u' \equiv pu \pmod{R}$. Every edge preserves D by Proposition 3.3. Exhaustive adjacency gives exactly three connected components: eight vertices at $D = 3$, twelve at $D = 5$, and six at $D = 15$. Thus combining these particular moves cannot turn this rejected shell into a passing shell. The graph does not restrict other cross-shell moves and is not a global counterexample.

Consequences for a least global counterexample. A least counterexample n must be prime: multiply a solution at any prime divisor by n/p . The prime 2 is solved by $(1, 2, 2)$. For $p \equiv 3 \pmod{4}$ the triple $((p+1)/2, (p+1)/2, p(p+1)/4)$ works. For $p \equiv 2 \pmod{3}$ use $(p, (p+1)/3, p(p+1)/3)$. A remaining prime is 1 $\pmod{12}$; if it is 13 $\pmod{24}$ then $a_3 = (p+3)/4$ is even and $u = 2$ passes the residual-three exterior gate. Therefore a least false prime is 1 $\pmod{24}$. Its first two shells obey

$$q \mid (p+3)/4 \implies q \equiv 1 \pmod{3},$$

$$q \mid (p+7)/4 \implies q \equiv 1, 2, 4 \pmod{7}.$$

The first assertion is the exact residual-three criterion. For the second, a_7 is even and the signed powers $2^{-1}, 1, 2$ fill the subgroup $\{1, 2, 4\}$ modulo seven. Any factor outside it makes the middle target -1 attainable using its one actual occurrence and one of those three powers. If every factor is inside, both original targets are outside. These are specializations of the foundation's first-two-shell theorems, not new source attributions [7].

The condition $p+1$ having a prime divisor $q \equiv 3 \pmod{4}$ also gives a witness: $R = q < p$, $r = s = 1$, $h = (p+q)/4$, and $q \mid p+1$. Thus a false prime has no such factor. Every further necessary condition proved below applies to every shell of the same false prime.

5 A sharpened positive collision criterion

Fix a shell and $G = (\mathbb{Z}/R\mathbb{Z})^\times$. Use the actual signed exponent box

$$\mathcal{B}_a^\pm = \prod_{q \mid a} \{-e_q, \dots, e_q\}.$$

Let $\mu(g)$ count its full fibres under $\beta \mapsto \prod q^{\beta_q} \pmod{R}$, and put

$$N = \prod_q (2e_q + 1), \quad E = \sum_g \mu(g)^2, \quad \mathcal{T} = \{-1, -p, -p^{-1}\} \subset G.$$

The set braces remove any coincident targets. Inversion of the box gives $\mu(g) = \mu(g^{-1})$. The channel targets on u/a are $-p^{-1}$ and -1 ; therefore avoiding both channels is exactly avoiding $\mathcal{T}$.

Theorem 5.1 (Integer collision-energy forcing). *Let $M = |G| - |\mathcal{T}|$. If $M > 0$ write $N = Mq + r$ with $0 \leq r < M$ and set*

$$L(N, M) = Mq^2 + r(2q + 1).$$

If both channels are empty, then $E \geq L(N, M)$. In particular $E < L(N, M)$ constructs a witness. If $M = 0$, a witness exists. The exact collision energy is

$$E = \sum_{\delta_q = -2e_q}^{2e_q} \mathbf{1}_{\prod q^{\delta_q} = 1 \pmod{R}} \prod_q (2e_q + 1 - |\delta_q|). \quad (\text{C})$$

Proof. A miss leaves at most M bins containing N occurrences. If two bin counts differ by at least two, moving one from the larger to the smaller strictly decreases the sum of squares. The minimum therefore has r bins of size $q + 1$ and $M - r$ bins of size q , giving L . For (C), E counts ordered pairs of exponent vectors with equal image. At a fixed difference δ_q , exactly $2e_q + 1 - |\delta_q|$ pairs occur in that coordinate; multiply and impose the one simultaneous residue equation. To extract a hit, retain an exponent vector in a target fibre. At -1 use M . At $-p^{-1}$ use E . At $-p$ negate the whole exponent vector first. In each case $u = \prod q^{e_q + \beta_q}$ remains inside the original divisor box and the canonical denominator formula returns the ordered witness. $\square$

A stronger exact symmetric minimum. If $1 \in \mathcal{T}$, the zero exponent vector is already a hit. Otherwise let s be the number of allowed nonidentity involutions and r the number of allowed unordered inverse pairs. The identity count is odd, each other involution count is even, and each inverse pair has equal counts. A miss consequently satisfies

$$E \geq L_{\text{sym}} := \min_{x_0 + \dots + x_s + y_1 + \dots + y_r = (N-1)/2} \left((1 + 2x_0)^2 + 4 \sum_{i=1}^s x_i^2 + 2 \sum_{j=1}^r y_j^2 \right),$$

with nonnegative integer variables. This finite minimum has an exact greedy algorithm: start from one and take $(N - 1)/2$ least available marginal costs from the increasing sequences $8, 16, 24, \dots$ for the identity, $4, 12, 20, \dots$ for each other involution, and $2, 6, 10, \dots$ for each inverse pair. Each selected prefix is feasible, and an exchange of an unnecessarily expensive selected marginal with a cheaper available one proves optimality. These constraints follow from the involution $\beta \mapsto -\beta$, whose sole fixed vector is zero.

Strict improvement at an inherited stress prime. For $p = 1201, a = 306, R = 23$, the factors are $2 \cdot 3^2 \cdot 17$. Here $N = 45, \mathcal{T} = \{9, 18, 22\}, M = 19$, and the exact energy is 107. Ordinary Cauchy–Schwarz gives only $19 \cdot 107 = 2033 > 2025$ and does not force a hit. The integer minimum is $12 \cdot 2^2 + 7 \cdot 3^2 = 111$; $L_{\text{sym}} = 111$ as well. Since $107 < 111$, the sharper theorem forces an actual target. The retained inverse returns $u = 17$, channel E , $(h, r, s, \kappa) = (17, 1, 18, 53)$ and $(306, 16218, 1082101)$. This is not new finite coverage of that prime; it is a strictly stronger exact sufficient criterion.

A hypothetical false prime must satisfy $E_a \geq L_{\text{sym},a}$ in *every* full shell. Equivalently the nontrivial Fourier mass of each multiplicity packet must be sufficiently large, since the usual finite Parseval identity gives $\sum_{\chi \neq 1} |\hat{\mu}(\chi)|^2 = |G|E - N^2$. No Fourier sign, density estimate, or subgroup-support approximation replaces the integer target fibre.

6 Exact maximal-cofactor angular forcing

Let $H \leq (\mathbb{Z}/R\mathbb{Z})^\times$ have index two with $-1 \notin H$. Let $A \mid a$, all of whose prime factors lie in H , and retain the *actual* finite anchor pairs

$$(n, d) : \quad nd \mid A, \quad \gcd(n, d) = 1.$$

Suppose their ratios cover H , an explicitly checkable condition on A . For $v \in H$, define the attained cost

$$c_A(v) = \min_{n/d=v \ (R)} d/n, \quad B_A = \max_{v \in H} c_A(v).$$

This is the preceding angular construction's exact finite data, not a universal assumption on the unknown ES prime.

Theorem 6.1 (Exact largest available outside divisor). *Write $a = Ab$ and assume $\gcd(a, R) = 1$. The largest divisor of b outside H is*

$$T_H(b) = \begin{cases} b, & b \notin H, \\ b/q_*, & b \in H \text{ and has a prime divisor outside } H, \\ 0, & \text{every prime divisor of } b \text{ lies in } H, \end{cases}$$

where q_* is the least actual outside prime factor. If $T_H(b) > 8B_A$, there is a middle state with $0 < r/s < 1/8$. More exactly, the anchor construction has such a state if and only if some actual outside divisor $t \mid b$ satisfies $t > 8c_A(-t^{-1})$.

Proof. If b is outside, b itself is the largest divisor. If $b \in H$ and t is outside, the complement b/t is outside, hence has an outside prime factor and is at least q_* . The bound $t \leq b/q_*$ is attained by that quotient. If there is no outside factor no divisor can be outside.

For an outside divisor t , choose a minimizing pair (n, d) for $-t^{-1} \in H$. Then $R \mid d + tn$ and $dtn \mid a$. Set

$$g = \gcd(d, tn), \quad r = d/g, \quad s = tn/g, \quad h = \frac{ag^2}{dtn}, \quad u = \frac{ad}{tn}, \quad \lambda = \frac{d + tn}{gR}.$$

Every coordinate is integral; g is a unit modulo R ; $a = hrs$, $u = hr^2$, and $\gcd(r, s) = 1$. Thus this is a middle state, and $r/s = c_A(-t^{-1})/t$. This proves both assertions. For a marked target (h, r, s) the complete inverse fibre consists of the anchor pairs for which $t = ds/(nr)$ is a positive divisor of b outside H and satisfies the stated congruence and inequality. Coprimality forces $r \mid d$ and then $g = d/r$. Thus all inverse data are explicit; forgetting (n, d, t) can create only the stated finite fibres. $\square$

The lower Koide boundary exceeds $1/8$: the rational bounds $\sqrt{3} > 1732/1000$, $\sqrt{2} < 1415/1000$, give $317/2415 > 1/8$. The radical bounds follow by squaring positive rationals. Thus the produced states lie in the stipulated lower chamber.

For $b \in H$ the preceding universal threshold $b > 64B_A^2$ is now replaced by the factor-sensitive threshold

$$b > 8B_A q_*.$$

The new theorem neither assumes the outside factor exists in every shell nor asserts monotonicity of D under an unspecified map.

Numerical certificate below the old sufficient threshold. Take $R = 19$, $A = 35$, and the quadratic residues $H = \{1, 4, 5, 6, 7, 9, 11, 16, 17\}$. The actual anchor box covers H and $B_A = 35$. At

$$p = 558721 = 140 \cdot 3991 - 19, \quad b = 3991 = 13 \cdot 307,$$

$b \in H$, $q_* = 13$ and $T_H(b) = 307 > 280$, whereas $3991 < 78400 = 64B_A^2$. The cost at the selected residue is 35, with $(n, d) = (1, 35)$. The theorem gives

$$(R, u, h, r, s, \lambda) = (19, 15925, 13, 35, 307, 18)$$

and ordered denominators

$$(139685, 40137399198, 4575924990).$$

The prime is 121 (mod 840) and its primality is certified by complete trial division. Three further examples and all minimizing pairs are in the JSON. These are illustrations of a stronger criterion, not claims of previously unknown ES coverage.

For a hypothetical false prime, every actual saturated anchor in every shell has *no* outside prime factor at all: without an angular restriction one outside divisor and the congruence construction already yield a middle state. For the stronger chamber-restricted counterfactual, the exact necessary inequalities are instead $t \leq 8c_A(-t^{-1})$ for every available outside t (for the $1/8$ subchamber). The two counterfactuals must not be conflated.

7 Transport through the marked geometric and lattice interfaces

7.1 Scale, Cayley coordinates, and all factor labels

For a rational atlas state $q = (a, Y, Z)$ retain

$$A = (-p, 4a, 4Y, 4Z), \quad e_3(A) = 0.$$

The cleared state (pD, Dq) has Cayley vector DA . Thus its projective point, all four face root triples, and their factor labels agree with those of the original rational candidate:

$$t_{ij} = -4A_i/A_j \quad (j \neq i), \quad \sum_{j \neq i} t_{ij} = 4.$$

The affine parameter p or the equivalent full scale must remain available. D can then be retained directly or recomputed from the rational denominators. It is not recoverable from branch labels alone. The face inverse is $A_j = -4A_i/t_{ij}$ with A_i retained.

For a flag (i, j, k, l) put $r = t_{ij}, s = t_{ik}, t = t_{il}$, $d = -\frac{1}{4}(r - s)(r - t)$ and $\eta = s - t$. The unnormalized factors

$$\ell = U - rV, \quad Q = -\frac{1}{4}(U - sV)(U - tV)$$

are retained for every flag, with resultant d . If $d \neq 0$, the normalized pair is $(\ell/d, dQ)$ and the inherited Fable source coordinates are

$$(d^{-1}, -r - d, 5d^2 + 3rd + \frac{1}{2}d^3).$$

The derivative, η , scale and flag are not deleted. If $d = 0$, the resultant-one chart is not invoked: the original factors and their multiplicity labels remain. In particular the rejected $M, u = a$ states are not thrown away to make the normalized chart look complete. The selected computations carry every one of 24 flags, including the zero-resultant strata, through the stated return.

The four-face resolvent from the preceding package is $b_0 = A_0A_1 + A_2A_3$ and its two matching analogues. Clearing gives $b_i(DA) = D^2b_i(A)$. This degree-two scale is retained. It does not identify the three resolvent values with the three denominator values.

7.2 A genuine failed rational candidate with integral geometric lifts

For $p = 241, a = 64, u = 1, E$, one has $R = 15, D = 3$ and

$$q = (64, 3904/3, 14701/3), \quad X = (192, 3904, 14701), \quad n = 723.$$

The integer X solves ES at 723, not at 241. Nevertheless both records have the same projective Cayley point and the same labelled face cubics. The certificate supplies all normalized simple factor routes for this failed candidate, not just routes for already successful states.

7.3 Leech fibres and the full Albert correction

Retain the preceding integral map $P : \Lambda \rightarrow \mathbb{Z}^3$ and its specified section $\sigma(t) = \sum t_i b_i$, where the numerator of b_0 is

$$(-3, 1, -1, 1, 1, -1, -1, 1, 1, 1, 1, 1, 1, -1, 1, 1, 1, 1, -1, -1, -1, -1, 1, 1)$$

and $b_1 = Cb_0, b_2 = C^2b_0$, each divided by $\sqrt{8}$. The three octads, code membership and identities $P(b_i) = e_i$ are checked exactly in the standalone package. The infinite section follows by linearity and closure of the classical Leech lattice.

The clearing denominator is recoverable even from the integral trace, provided the original prime is retained:

$$D = \frac{4X_0X_1X_2}{p \sum_{i < j} X_iX_j}.$$

It then recovers $q = X/D$, a , R , and u by the candidate inverse. The channel is distinguished by $v_p(Y) = 0$ in E and $v_p(Y) > 0$ in M . Thus the map from canonical candidates to their ordered cleared triples, and hence to their chosen Leech lifts, is injective for fixed p . The complete scaled-candidate lift is

$$(\alpha, k) \mapsto \lambda_\Lambda = \sigma(Dq) + k, \quad k \in \ker P.$$

Its inverse uses $q = P\lambda_\Lambda/D$, the retained p, R, c and the candidate inverse, then $k = \lambda_\Lambda - \sigma(Dq)$. These are maps on the explicitly specified image, not on arbitrary unmarked lattice vectors. The supplied tests use $k = 0$ but retain the full fibre theorem. The all-octad inverse is checked on these vectors too; its nonzero-syndrome classes are not silently admitted.

The signed Golay–Wilson map in `lineage/aligned_isometry.json` identifies the full numerator vector with ordered octonions z_0, z_1, z_2 . Its explicit signed inverse is checked. The previously established infinite-lattice image theorem is inherited, not newly audited here. The checker verifies the signed dictionary against the pinned matrix on all 576 entries, tests the actual joint Wilson membership conditions on each returned state, and computes the full octonion cubic using the retained Fano multiplication convention [1]. For $X = P\lambda_\Lambda$, $n_i = N(z_i)$ and $\nu = 2\Re((z_0 z_1) z_2)$, retain

$$\hat{X} = \begin{pmatrix} X_0 & z_2 & \bar{z}_1 \\ \bar{z}_2 & X_1 & z_0 \\ z_1 & \bar{z}_0 & X_2 \end{pmatrix}.$$

Then the earlier exact identity becomes

$$4N_J(\hat{X}) - pS_J(\hat{X}) = 2p\delta_\alpha \sum_{i < j} X_i X_j + \sum_i (p - 4X_i)n_i + 4\nu.$$

The seven geometric test states include the numerical values of N_J, S_J, ν , and both sides of this identity. The diagonal denominator defect and the off-diagonal cubic correction are separate terms. Neither an arbitrary lattice isometry nor a split-Tits transformation is declared to preserve the marked compact cubic.

For any genuine bijection of retained states f , multiplication by $\delta \circ f^{-1}$ is conjugate to multiplication by δ on the original function algebra. Thus success kernels and negative inverse certificates transport exactly. A many-to-one projection must retain its fibre, and a map changing the arithmetic state must recompute D ; functorial notation alone does not prove an improvement.

8 The infinite trace-space counterfactual and its analytic boundary

Since $P^{-1}(t) = \sigma(t) + K$, $K = \ker P$, the positive trace space is explicitly $\mathbb{Z}_{>0}^3 \times K$. Multiplication by $F_p(P\lambda)^2$ has a nonzero kernel exactly when ES holds at p . Under the false-prime hypothesis all values are positive integers, so its pointwise inverse $1/F_p(P\lambda)^2$ exists and is bounded by one. Under the true-prime alternative every successful trace gives an entire rank-21 lattice fibre of zeros. More precisely, on $\ell^2(\{\lambda \in \Lambda : P\lambda \in \mathbb{Z}_{>0}^3\})$ with counting measure, its domain is

$$\mathcal{D}(\mathcal{H}_p) = \{f : \sum_\lambda F_p(P\lambda)^4 |f(\lambda)|^2 < \infty\}.$$

Coordinatewise multiplication by the nonnegative real values $F_p(P\lambda)^2$ is self-adjoint on this maximal domain: the adjoint condition tested against each coordinate vector forces the same multiplier and the same square-summability requirement. In the false case its inverse is the everywhere-defined multiplier $F_p(P\lambda)^{-2}$, with norm at most one, and $\mathcal{H}_p \geq I$. In the true case its zero eigenspace is infinite-dimensional, since an entire translate of the rank-21 kernel is present. This is an exact operator encoding of the arithmetic alternative, not an independently constructed differential operator or an asserted analytic solution of the conjecture.

The same alternative is expressed by the previously marked theta series:

$$\mathcal{Z}_p(q) = \sum_{\substack{t \in \mathbb{Z}_{>0}^3 \\ F_p(t)=0}} q^{\|t\|^2/8} \Theta_{K+r(t)}(q), \quad 0 < q < 1.$$

Every summand is positive, so $\mathcal{Z}_p(q) = 0$ exactly in the false case. This identifies the analytic quantity to control; it does not calculate it independently of arithmetic occupancy.

There is a useful external restriction on any genuine counterexample. Bright–Loughran’s Theorem 1.8 proves that the positive-real/integral-adic Brauer–Manin set is nonempty for every parameter [6]. A false prime would therefore have a nonempty such adelic set but no global positive integer witness. This rules out one proposed emptiness explanation, not all possible arithmetic obstructions. It also does not turn local compatibility inside different exponent vectors into one simultaneous bounded lift.

9 The two research directions as one program

The positive track tries to construct a zero of δ_α . The negative track assumes δ is a unit and derives its mandatory arithmetic, concentration, geometric and integral-return properties. A contradiction must violate one of those properties by a proved calculation. The supplied positive tests are Theorem 5.1 and Theorem 6.1; the exact same-shell obstruction is Proposition 3.3.

A next useful theorem would prove that every full atlas has a shell whose integer energy falls below its counterfactual minimum, or construct a well-founded cross-shell move that strictly lowers D until $D = 1$. No such universal decrease is assumed. The inverse certificate and the restricted examples make a proposed move testable on failures as well as successes. A proof of either universal assertion would finish the positive prime cases; the present tranche proves neither assertion.

10 Certificates, checks, and nonclaims

The standard-library checker exhausts every full canonical candidate at the following seven primes:

p	5	13	37	97	241	1201	2521
leaves	12	64	312	1216	3964	29600	73060

For 1201 the full counts are $8E + 8M$, not the $7E + 8M$ of the restricted $R < p$ atlas; at 2521 they are $7E + 6M$, not $6E + 6M$. These are marked first-denominator counts, not sorted-triple counts. All assertions about rational denominators, scaling, inverse coordinates and repeated-root strata are checked on every candidate.

The two negative examples are labelled *restricted*; the validator rejects promoting them to full certificates. It also rejects a removed leaf, an incorrect clearing denominator, and a fabricated full negative certificate at $p = 5$. The exact energy autocorrelations are recomputed independently of their histogram squares. The largest-outside-divisor formula is compared with exhaustive divisor enumeration for every unit cofactor $b \leq 3000$ at $R = 19$. All 24 factor labels are retained for seven selected rational or integral states, including one repeated-root state. No actual full negative certificate is found.

Run

```
python dual_es.py --self-test --out certificates
python -O dual_es.py --self-test --out certificates_optimized
```

The optimized run has identical mathematical outputs. Use `-prime p` to produce a witness or a fully verified negative certificate, and `-verify-negative file.json` to check a proposed full negative certificate. Trial division and complete enumeration are finite but are not advertised

as polynomial in $\log p$. The source Giuga program was read, not fully executed; its expensive power-sum and symbolic Cayley–Dickson suites are not part of these receipts. No new numerical verification bound, independent reviewer, formal Lean build, or proof of the global conjecture is claimed.

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